Section 5.4

The Distribution of the Sample Mean

Recall: Random Samples

- A **random sample** $X_1, \ldots, X_n$ satisfies:
  - the X_i's are independent;
  - every X_i has the same distribution.

- The **sample mean** is the statistic

$$\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

- $\overline{X}$ is a random variable, so it has a distribution: the **sampling distribution** of $\overline{X}$.

- Today: the mean, variance and shape of this distribution.

Mean and Variance of $\overline{X}$

- Let $X_1, \ldots, X_n$ be a random sample from a distribution with mean μ and standard deviation σ. Then

$$E(\overline{X}) = \mu_{\overline{X}} = \mu \qquad V(\overline{X}) = \sigma^2_{\overline{X}} = \frac{\sigma^2}{n} \qquad \sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}$$

- For the **sample total** $T_0 = X_1 + \cdots + X_n$:

$$E(T_0) = n\mu \qquad V(T_0) = n\sigma^2 \qquad \sigma_{T_0} = \sqrt{n}\,\sigma$$

- $\overline{X}$ is centered at the population mean μ.
- The spread of $\overline{X}$ shrinks as n grows.
- $\sigma_{\overline{X}} = \sigma/\sqrt{n}$ is called the **standard error** of the mean.

Example 1: The fill weight of a cereal box has mean $\mu = 500$ g and standard deviation $\sigma = 10$ g. A random sample of $n = 25$ boxes is selected.

a) Find $E(\overline{X})$, $V(\overline{X})$ and $\sigma_{\overline{X}}$.

b) Find $E(T_0)$, $V(T_0)$ and σ_{T_0}, where T_0 is the total weight of the 25 boxes.

c) What is $\sigma_{\overline{X}}$ if the sample size is increased to $n = 100$?

The Case of a Normal Population

- Let $X_1, \ldots, X_n$ be a random sample from a normal distribution, $X_i \sim N(\mu, \sigma^2)$. Then, for **every** n,

$$\overline{X} \sim N\left(\mu,\ \frac{\sigma^2}{n}\right) \qquad T_0 \sim N\left(n\mu,\ n\sigma^2\right)$$

- Probabilities are found by standardizing:

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1) \qquad Z = \frac{T_0 - n\mu}{\sqrt{n}\,\sigma} \sim N(0,1)$$

- This result is exact: no large-sample condition is needed.

Example 2: The time (in minutes) a worker needs to assemble a part is normally distributed with mean 12 and standard deviation 2. A random sample of $n = 16$ workers is observed.

a) Find $P(\overline{X} \leq 12.5)$.

b) Find $P(11.5 \leq \overline{X} \leq 12.5)$.

c) Find the probability that the total time of the 16 workers is more than 200 minutes.

The Central Limit Theorem

- Let $X_1, \ldots, X_n$ be a random sample from **any** distribution with mean μ and variance σ^2. If n is sufficiently large, then

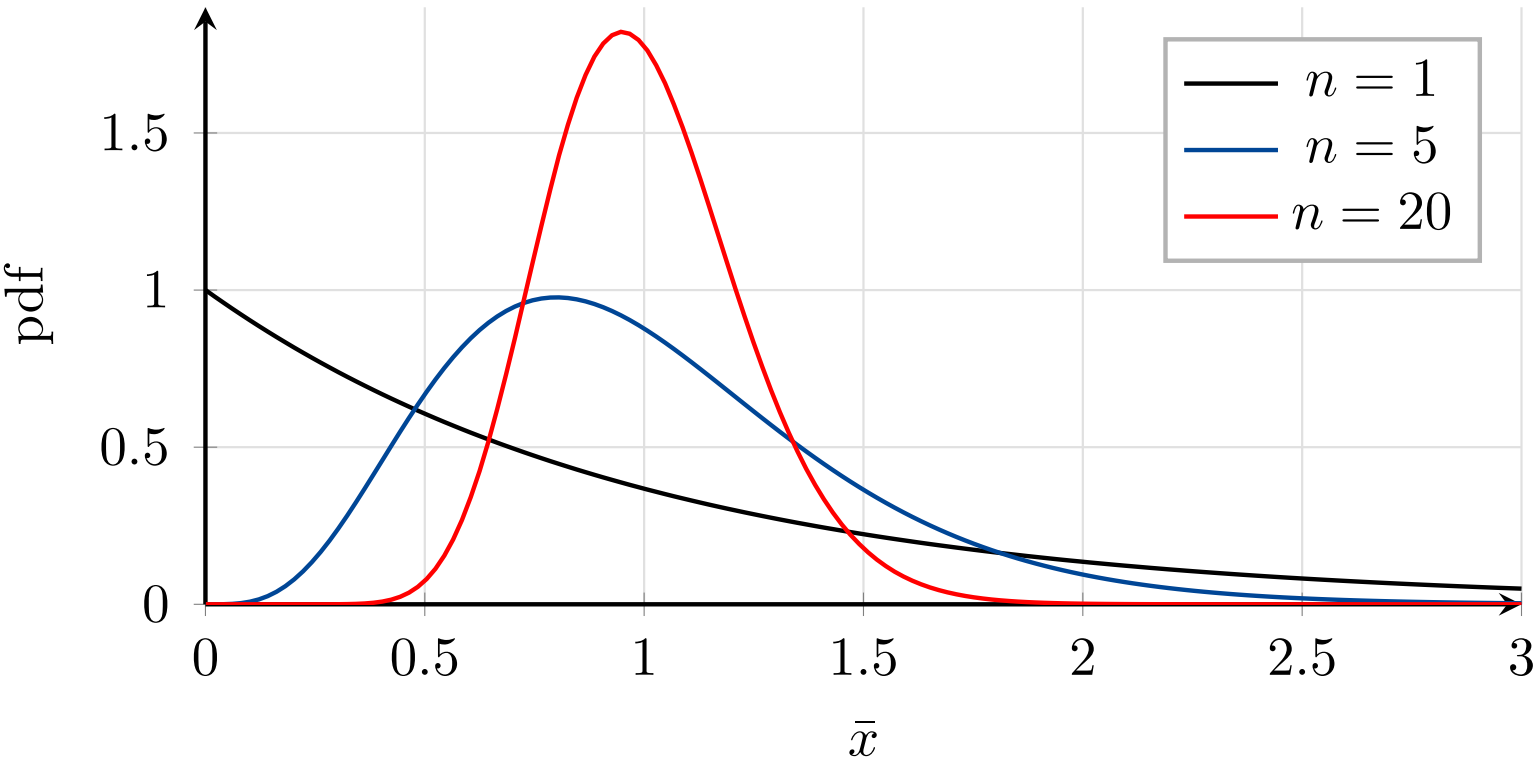
$$\overline{X} \text{ is approximately } N\left(\mu, \frac{\sigma^2}{n}\right)$$

$$T_0 \text{ is approximately } N\left(n\mu, n\sigma^2\right)$$

- The larger n is, the better the approximation.
- Rule of thumb: the **Central Limit Theorem** (CLT) may be used if $n > 30$.
- If the population is already normal, $\overline{X}$ is exactly normal for every n.

The CLT: An Illustration

- Population: exponential with $\mu = 1$ (strongly skewed).
- The pdf of $\overline{X}$ for $n = 1,\ 5,\ 20$:



- As n grows, the pdf of $\overline{X}$ gets narrower and more bell-shaped around $\mu = 1$.

Example 3: The length of a customer service call has mean 4 minutes and standard deviation 3 minutes. The distribution is skewed to the right. A random sample of $n = 36$ calls is observed.

a) Find the approximate probability that the sample mean call length is more than 4.5 minutes.

b) Find the approximate probability that the total length of the 36 calls is at most 162 minutes.

c) Why can we use the normal distribution here, even though call lengths are not normal?

Example 4: The number of flaws X on a sheet of glass has pmf

x	0	1	2
$p(x)$	0.5	0.3	0.2

A random sample of $n = 100$ sheets is inspected.

a) Find μ and σ^2.

b) Find the approximate probability that the sample mean number of flaws is at most 0.8.

Other Applications of the CLT

- Let $X \sim \text{Bin}(n, p)$. Then $X = X_1 + \cdots + X_n$, where

$$X_i = \begin{cases} 1 & \text{if trial } i \text{ is a success} \\ 0 & \text{if trial } i \text{ is a failure} \end{cases}$$

  so X is a sample total T_0, with $\mu = p$ and $\sigma^2 = p(1-p)$.
- By the CLT, for large n:

$$X \text{ is approximately } N\Big(np,\ np(1-p)\Big)$$

- The sample proportion $\hat{p} = X/n$ is a sample mean, so it is approximately $N\Big(p,\ p(1-p)/n\Big)$.
- Rule of thumb: use this if $np \geq 10$ and $n(1-p) \geq 10$.

Summary

Population	Sample size	Distribution of $\overline{X}$
any	any n	$E(\overline{X}) = \mu, \quad V(\overline{X}) = \sigma^2/n$
normal	any n	exactly $N(\mu, \sigma^2/n)$
not normal	$n > 30$	approximately $N(\mu, \sigma^2/n)$ (CLT)

- To find probabilities, standardize: $Z = \dfrac{\overline{X} - \mu}{\sigma/\sqrt{n}}$.
- For the total, use T_0 with mean $n\mu$ and standard deviation $\sqrt{n}\,\sigma$.